Project - CSIS4260

Data Analytics Project — Machine Learning Classification

Team Members

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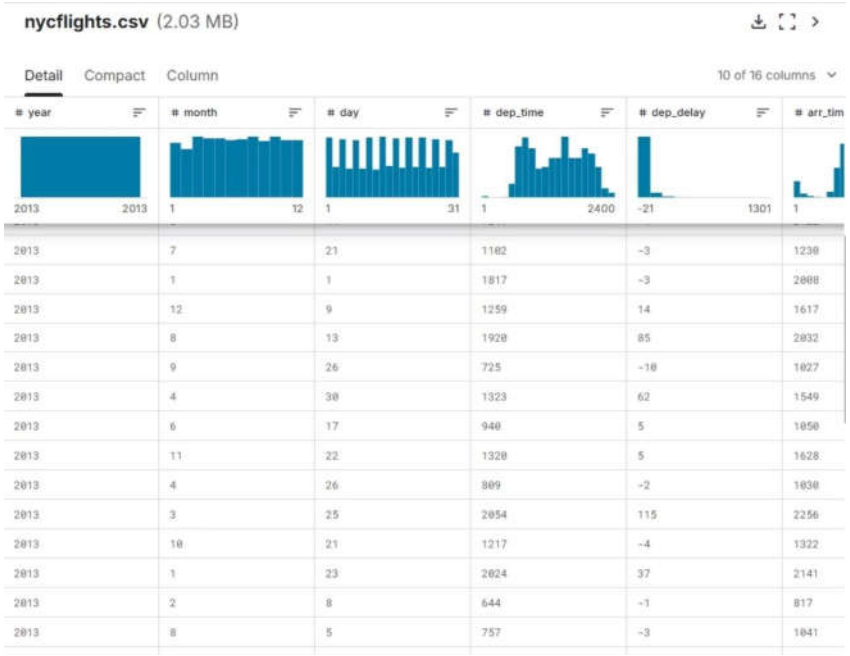
Project Overview

The main purpose of this project is to analyze flight delay data and use machine learning to predict **arrival delay**. Flight delays are an important real-world problem because they affect passengers, airlines, airport operations, and scheduling efficiency.

Our objectives are:

- To understand the flight dataset and weather dataset,
- To clean and prepare the data for analysis,
- To create useful features for prediction,
- To compare different machine learning models,
- And finally, to identify which factors have the strongest impact on arrival delay.

Data Sources



nycflights.csv (2.03 MB)

Detail Compact Column 10 of 16 columns

# year	# month	# day	# dep_time	# dep_delay	# arr_tim
2013 – 2013	1 – 12	1 – 31	1 – 2400	-21 – 1301	1
2013	7	21	1102	-3	1230
2013	1	1	1817	-3	2008
2013	12	9	1259	14	1617
2013	8	13	1920	85	2032
2013	9	26	725	-10	1027
2013	4	30	1323	62	1549
2013	6	17	940	5	1050
2013	11	22	1320	5	1628
2013	4	26	809	-2	1030
2013	3	25	2054	115	2256
2013	10	21	1217	-4	1322
2013	1	23	2024	37	2141
2013	2	8	644	-1	817
2013	8	5	757	-3	1041

Primary Source

The first source is the **NYC flights dataset**, which contains information about flights such as departure time, departure delay, arrival time, arrival delay, carrier, origin airport, destination, air time, and distance...

Second Source:

weather dataset, which was merged with the flight data.
This added weather variables such as temperature, precipitation, rain, snowfall, wind speed, wind gusts, and weather condition code.

After combining

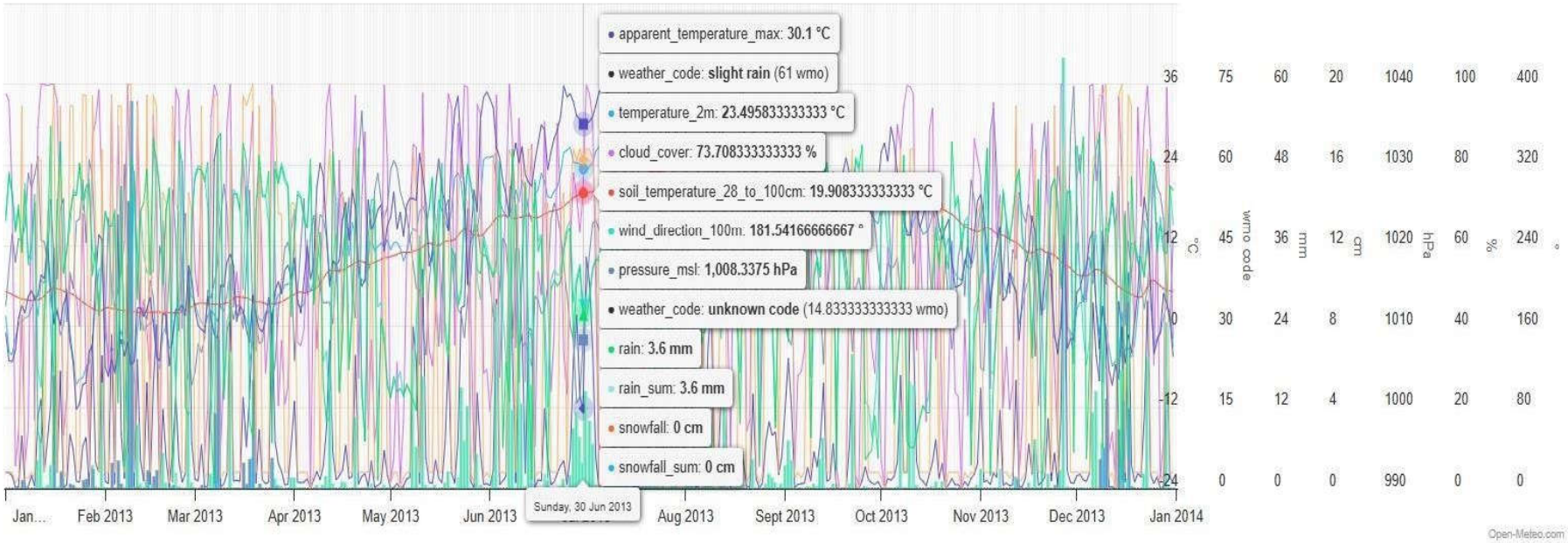
After combining both sources, our enriched dataset contains **32,735 rows and 29 columns**.
This gives us a stronger dataset because flight delays are often influenced not only by airline and schedule factors, but also by weather conditions.

API Response

Preview: Chart & URL | Python | TypeScript | Swift | Other

40.74°N -74.04°E 51m above sea level

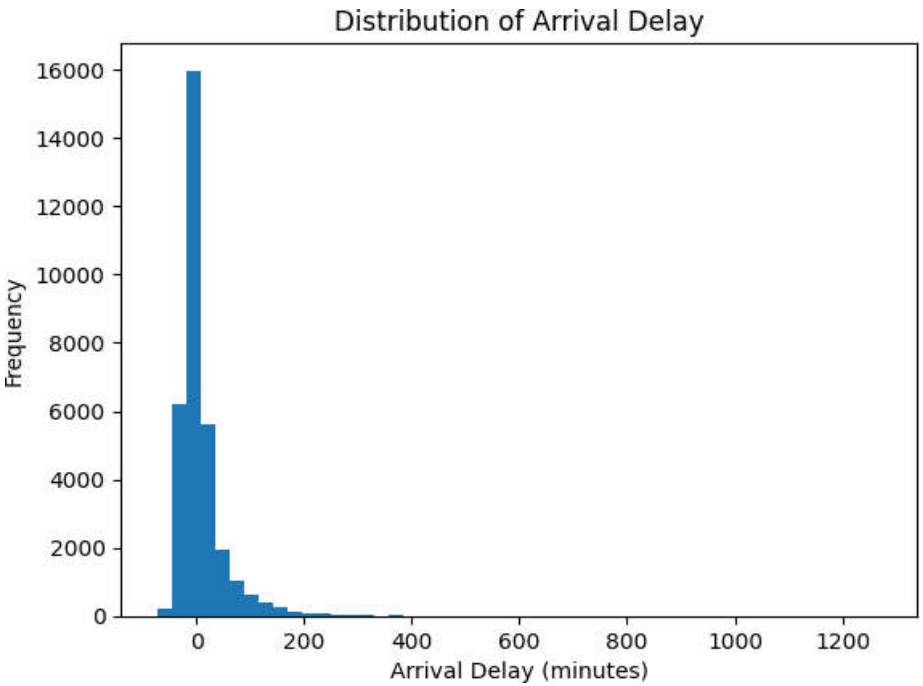
Generated in 10.45ms, downloaded in 575ms, time in GMT+0

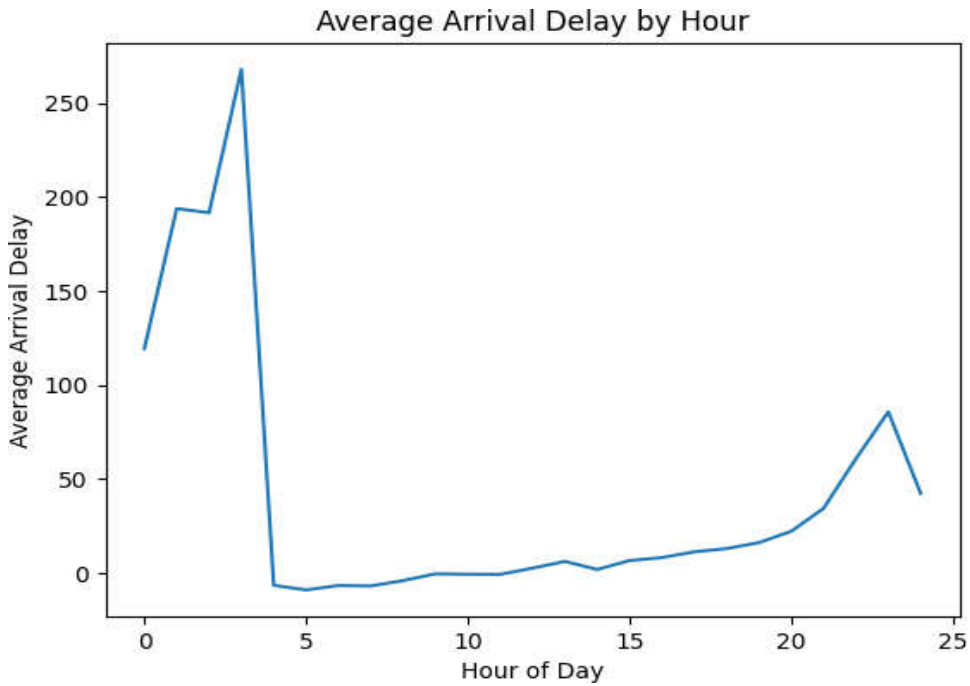


Download XLSX | Download CSV

API URL (Open in new tab or copy this URL into your application)

EDA (EXPLORE DATA ANALYSIS)

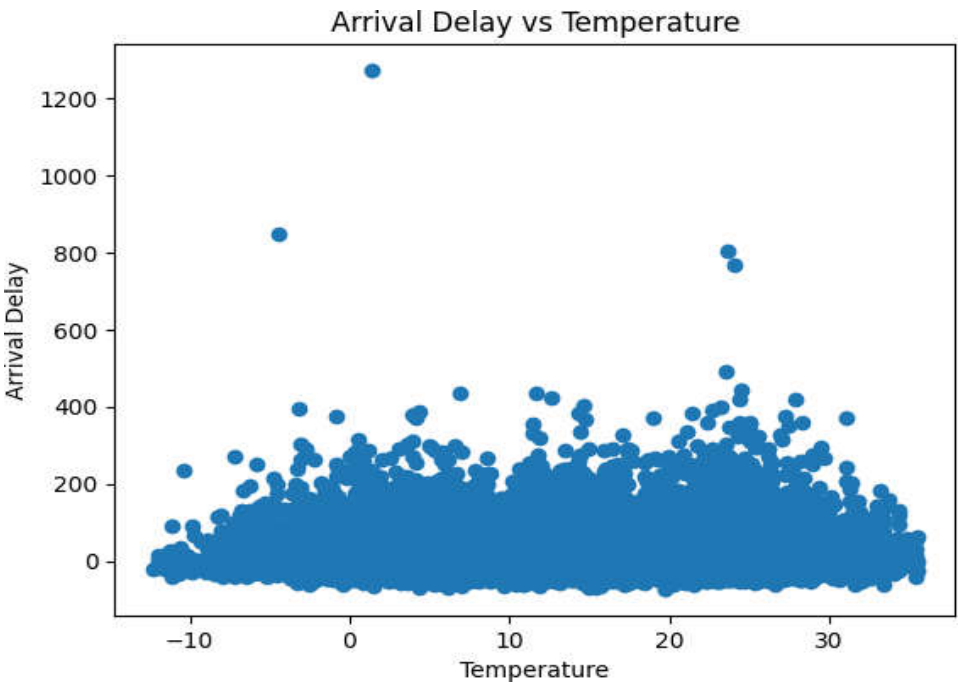




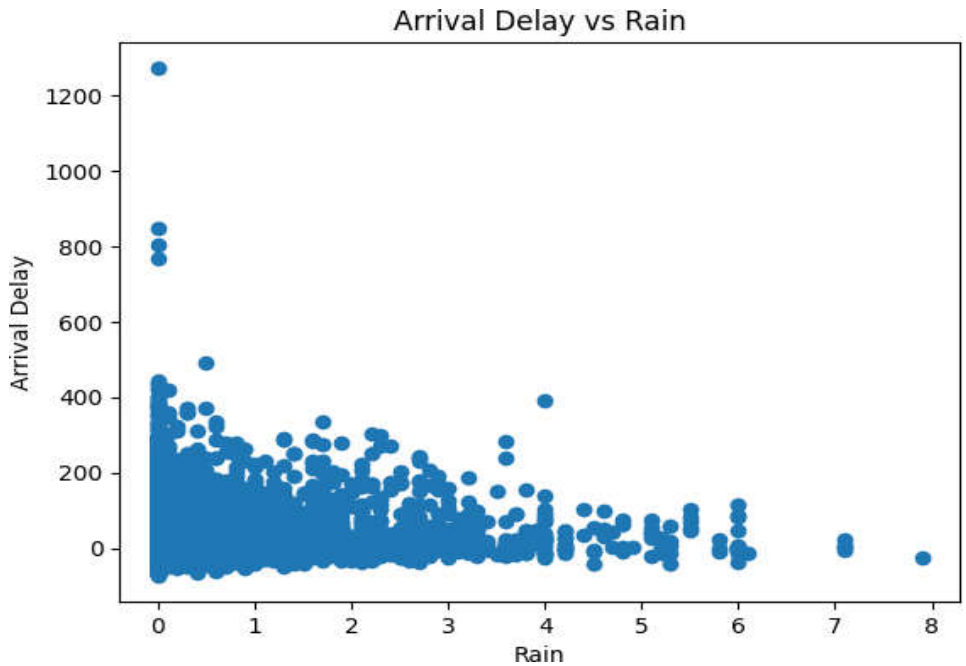
The chart gives information about distribution of arrival delay

The graph shows the average arrival delay by hour per day

EDA (EXPLORE DATA ANALYSIS) continue:



Scatter plot between Arrival Delay and Temperature



Scatter plot between Arrival Delay and Rain

Features Selection

Target: arr_delay (arrival delay)

add

Basic schedule features

- month
- day
- hour
- minute

Flight identity / route features

- carrier
- origin
- destination
- route-based information

Weather features

- temperature
- precipitation
- rain
- snowfall
- wind speed
- wind gusts

remove

- dep_delay
- air_time

Feature Engineering

⇒ **to improve model performance**

⇒ **help model learn patterns and improve accuracy**

Historical delay:

- average historical delay for the same route
- average historical delay for the same carrier
- average historical delay for the same carrier-route combination

Weather risk score:

- Check each raw weather column
- Turn each bad condition into a 0/1 flag
- Add all those flags together
- Use the final total as a summary of weather severity

Modeling Approach

Problem type: Regression (predict arrival delay)

Models

- Linear Regression (baseline)
- Random Forest (nonlinear model)
- HistGradientBoosting (tuned)

Pipeline

- Missing value imputation
- One-hot encoding for categorical features
- Model training using scikit-learn pipeline

Evaluation

- Chronological split (train / validation / test)
- Metrics:
  - RMSE: penalizes large errors
  - MAE: average prediction error
  - R^2 score: proportion of variance explained

Model Selection

- Best model chosen based on lowest RMSE

Model Evaluation (Test Set)

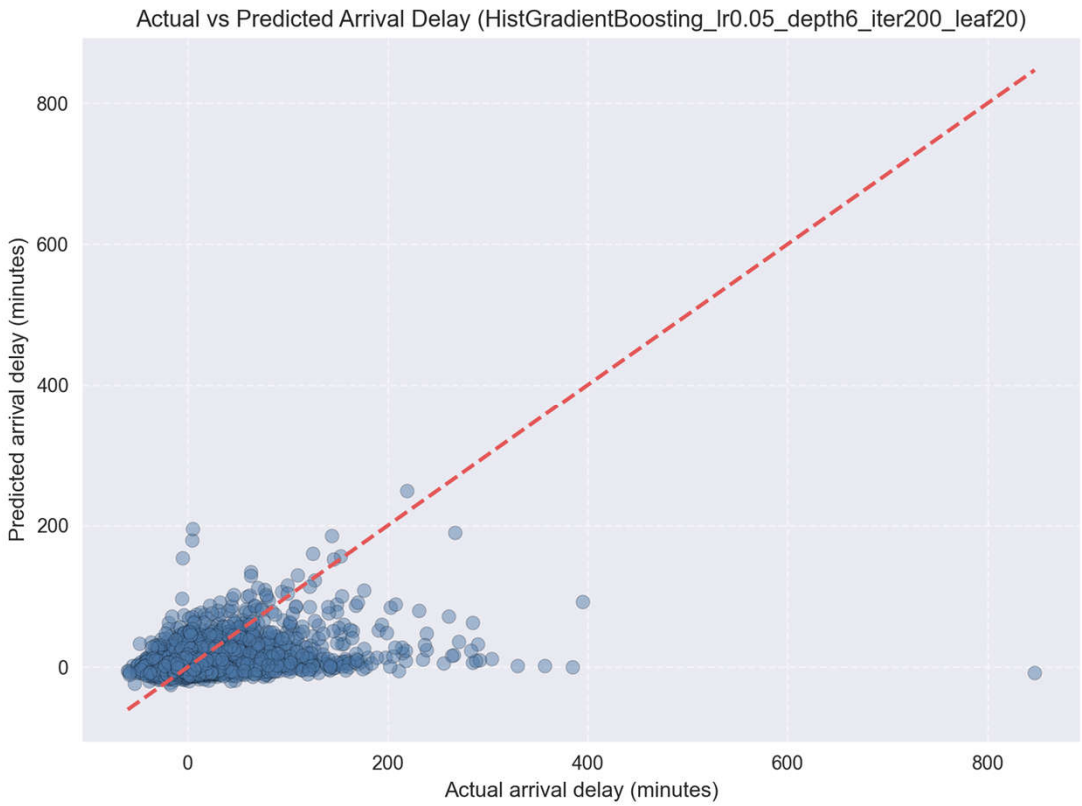
- Performance on unseen test data comparing baseline against non-linear estimators.

Key Takeaway:

- Tree-based models outperform linear models in capturing nonlinear patterns.
- HistGradientBoosting captures complex nonlinear relationships in delay patterns more effectively.

Model Architecture	RMSE (mins) ↓	MAE (mins) ↓	R^2 ↑
Linear Regression	48.55	32.5	-0.361
Random Forest	**39.92**	**24.51**	0.08
HistGradientBoosting ★ **BEST**	**38.54**	**22.35**	**0.142**

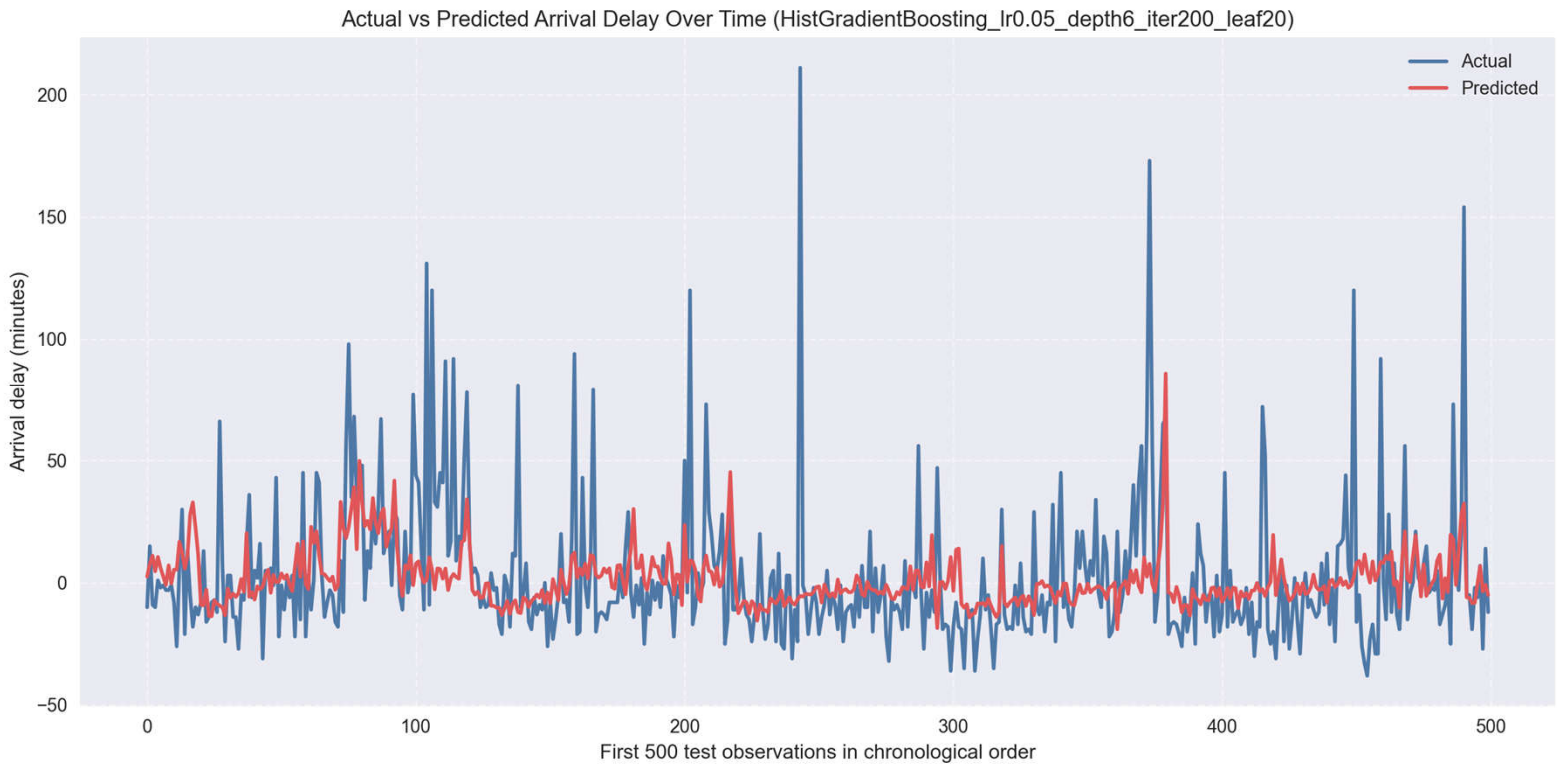
PREDICTION PERFORMANCE



Actual vs Predicted Delay

- Most predictions cluster near small delays

PREDICTION PERFORMANCE



Prediction vs Actual over Time

- Model captures overall delay trends
- Good performance on normal delays
- Underestimates extreme delays (outliers) due to rare events

Flight Delay Prediction Dashbo

localhost:8501

Ask Gemini

Deploy

Model Info

Best model: HistGradientBoosting_lr0.05_depth6_iter200_leaf20

Artifacts expected:

```
outputs/best_model.pkl
outputs/model_results.csv
nycflights_with_weather.csv
```

✈️ Flight Arrival Delay Prediction Dashboard

Predict arrival delay using the trained model and flight + weather inputs.

Enter Flight Details

Year: 2013

Month: 1

Day: 15

Departure Hour: 12

Departure Minute: 0

Carrier: 9E

Origin: EWR

Destination: ABQ

Weather Inputs

Temperature (°C): 30.00

Rain: 222.00

Wind Speed: 12.73

Precipitation: 0.11

Snowfall: 0.00

Wind Gusts: 27.24

Weather Condition: Snow

Predict Delay

Model Performance

	model	split	rmse	mae	r2
0	HistGradientBoosting_lr0.05_depth6_iter200_leaf20	test	38.542	22.3457	0.1424
1	Random Forest	test	39.924	24.5058	0.0798
2	Linear Regression	test	48.5487	32.4991	-0.3607

Quick Notes

- This app loads the trained `best_model.pkl`.
- Historical route/carrier features are estimated from the training dataset.
- For unseen routes, the app falls back to global average delay values.
- For best results, keep `app.py`, `nycflights_with_weather.csv`, and the `outputs` folder together.

Download model results CSV

Conclusion

- Successfully combined **flight data and weather data** into one dataset.
- Performed **data cleaning and feature engineering** to prepare the data.
- Built and compared **two models: Linear Regression and Random Forest**
- **Linear Regression performed better:**
- RMSE = **14.56.**
- Random Forest RMSE = **17.61.**
- Indicates a **strong linear relationship**, especially with **departure delay.**
- Both **flight features and weather features** influence arrival delay.
- Machine learning can effectively **predict real-world problems like flight delays.**